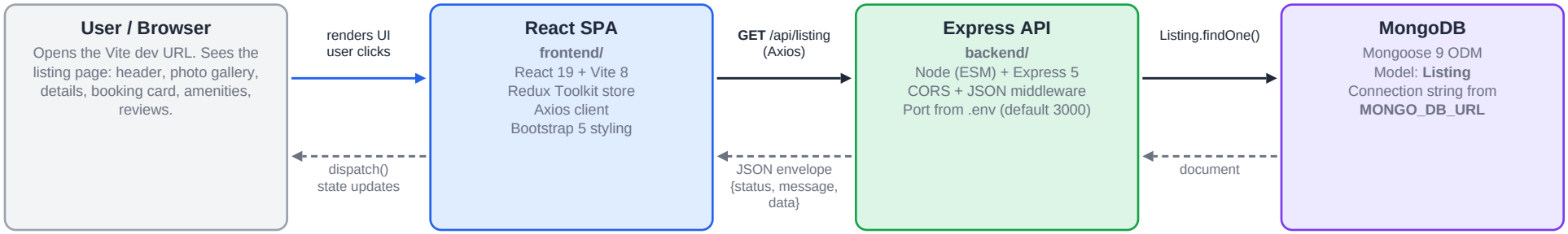
Airbnb Clone - Architecture Diagram

Full-stack listing page: React + Redux Toolkit SPA | Express 5 REST API | MongoDB via Mongoose

System overview



Runtime wiring: the SPA calls `http://localhost:3000/api/listing` (hard-coded in `listingSlice.js`). The API only accepts cross-origin requests from `CORS_URL`, so that value must equal the Vite dev origin. Backend .env keys: `PORT, MONGO_DB_URL, CORS_URL`.

Stack at a glance

Frontend (frontend/package.json)

- **react / react-dom** ^19.2.8
- **vite** ^8.3.0 + @vitejs/plugin-react
- **@reduxjs/toolkit** ^2.12.0, **react-redux** ^9.3.0
- **axios** ^1.20.0 (HTTP)
- **bootstrap** ^5.3.8 (layout & components)
- **react-router** ^8.4.0 - installed, not used yet (no router in main.jsx)
- ESLint 10 + react-hooks / react-refresh plugins

Backend (backend/package.json)

- **express** ^5.2.1
- **mongoose** ^9.10.2
- **cors** ^2.8.6
- **dotenv** ^18.0.4
- **nodemon** ^3.1.14 (dev)
- "type": "module" (ES modules)
- Dev script: `nodemon sever.js`
- Layered folders: config / routes / controllers / models

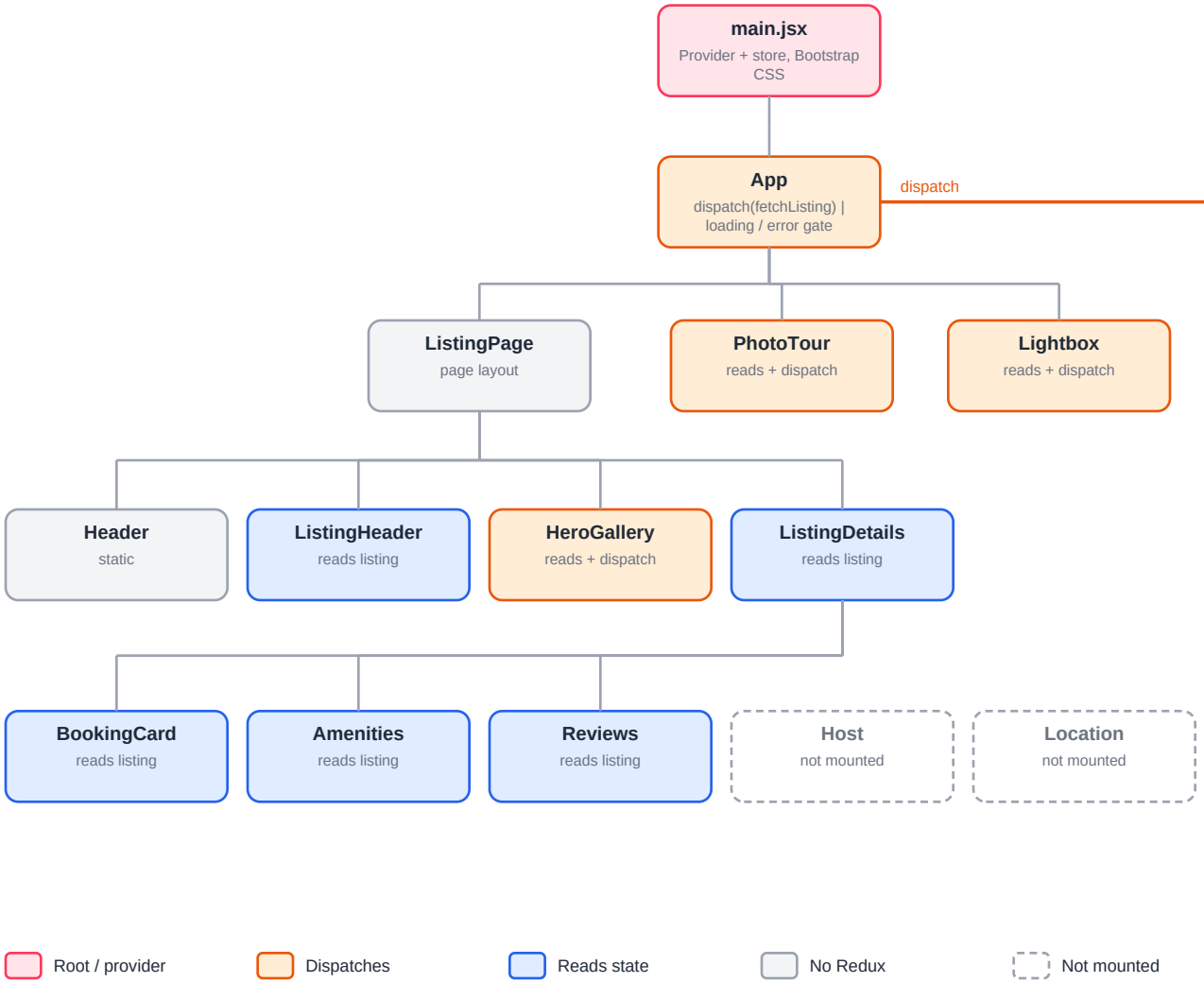
Repository layout

```
airbnb-clone/
  backend/
    sever.js        (entry)
    .env
    src/config/db.js
    src/routes/listingRoutes.js
    src/controllers/listingController.js
    src/models/listingModel.js
  frontend/
    src/main.jsx  App.jsx
    src/pages/ListingPage.jsx
    src/components/  (11 files)
    src/redux/store.js
    src/redux/slices/listingSlice.js
  package.json  (root)
```

Frontend Architecture

React component tree, Redux Toolkit store and how components read / dispatch

Component tree (frontend/src)



Redux store - state.listing

Initial state

- `listing`: null -> Listing document
- `loading`: false
- `error`: null
- `currentPhoto`: 0 (index)
- `showPhotoTour`: false
- `showLightbox`: false

Async thunk

fetchListing
axios.get('.../api/listing')
pending -> loading = true
fulfilled -> listing = data
rejected -> error = message

Reducers / actions

- `openPhotoTour()` — show grid modal
- `closePhotoTour()` — hide grid modal
- `openLightbox(i)` — set photo, show viewer
- `closeLightbox()` — hide viewer
- `nextPhoto()` — (i + 1) mod count
- `previousPhoto()` — (i - 1) mod count

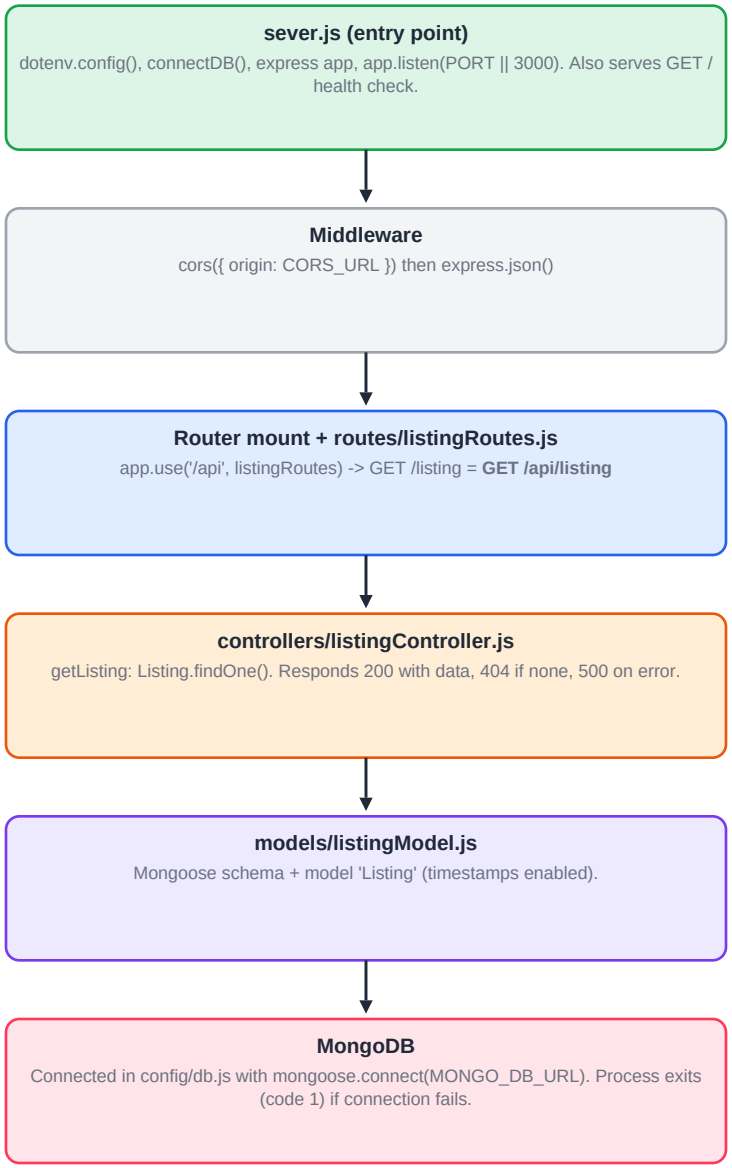
Wired in `redux/store.js` via `configureStore({ reducer: { listing } })`.

Data flow: App triggers the fetch once on mount. Every other component is a pure view over `state.listing.listing`; only UI toggles (photo tour, lightbox, current photo) are written back through actions.

Backend Architecture

Express request pipeline and MongoDB data model

Request pipeline (backend/)



Client request enters at the top; the JSON response travels back up the same chain.

Listing schema (one document = one property)

Field	Type	Notes
`title`	`String`	required
`location`	`String`	required
`rating`	`Number`	default 0
`reviews`	`Number`	default 0 (count)
`guests`	`Number`	required
`bedrooms`	`Number`	required
`beds`	`Number`	required
`bathrooms`	`Number`	required
`price`	`Number`	required
`host`	`String`	required (name only)
`description`	`String`	required
`amenities`	`[String]`	list of labels
`photos`	`[{ id: Number, image: String }]`	gallery images
`createdAt / updatedAt`	`Date`	auto (timestamps)

API contract

GET /api/listing

- **200** {status: true, message, data: Listing}
- **404** {status: false, message: 'Listing not found'}
- **500** {status: false, message, error}
- No path params: always returns the first document.

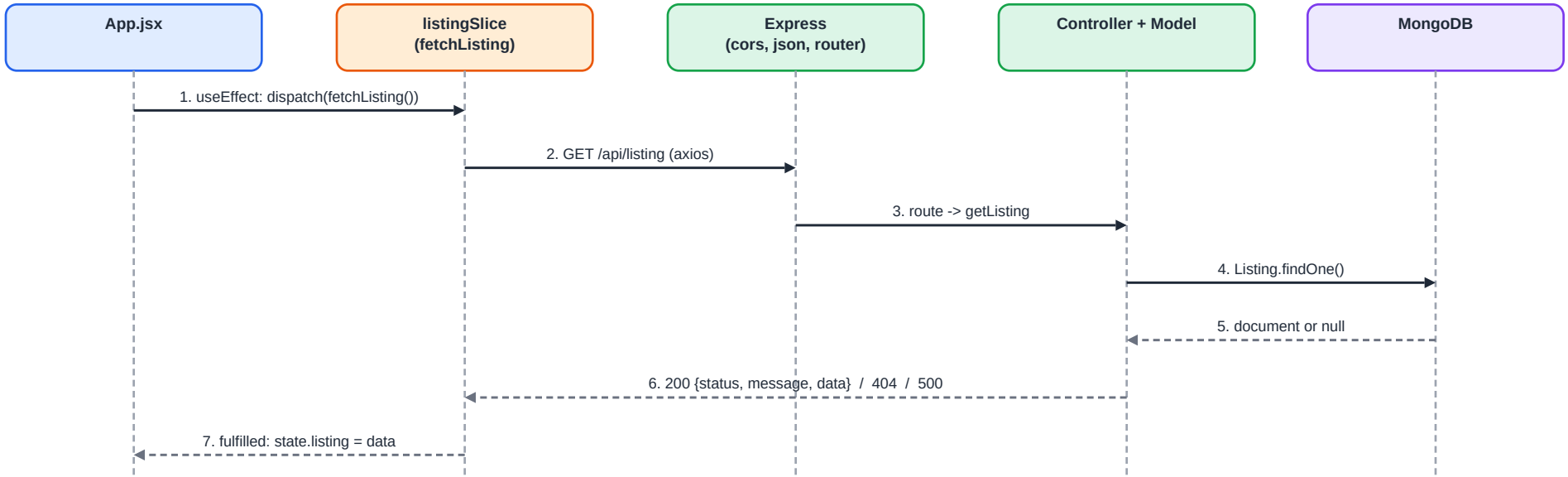
Environment (backend/.env)

- `PORT` - listen port (default 3000)
- `MONGO_DB_URL` - MongoDB connection string
- `CORS_URL` - allowed browser origin
- Values are not reproduced in this document.

Runtime Flow & Observations

What happens on page load, plus things worth tightening as the project grows

Page-load sequence



Meanwhile `fetchListing.pending` sets `loading = true` (spinner). After step 7 the tree re-renders from `useSelector`. On any failure `rejected` stores `error` and App shows the red alert.

Observations & suggested next steps

• **Entry file name:** the backend entry is `sever.js` (likely meant `server.js`); package.json also lists `main: index.js`, which does not exist.

• **Hard-coded API URL:** `listingSlice.js` uses `http://localhost:3000`. Move it to a Vite env var (e.g. `VITE_API_URL`) before deploying.

• **Single-listing model:** the API has one route and returns `findOne()`. A `/listings/:id` route would unlock multiple properties.

• **Unused pieces:** `react-router` is installed but there is no router; `Host.jsx` and `Location.jsx` are not mounted; `src/data/listing.js` is empty.

• **Secrets in the archive:** `backend/.env` is included in the zip and no backend .gitignore appears. Make sure it is ignored by git and rotate the DB credentials if it was shared.

• **Not yet built:** users/auth, host model (host is a plain string), booking flow (BookingCard only reads listing data), and individual review documents.

• **Root package.json** repeats frontend dependencies (redux, axios, bootstrap, router); it can probably be removed or turned into a workspace file.